Derivations of RC LPF and HPF Transfer Functions

The following section will be used as an exercise deriving the transfer functions of second order RC Low Pass and High Pass filters. The filters that make up the graphical EQ will be a combination of cascaded second order high pass and low pass filters coupled using operational amplifiers as buffers to prevent the first filter from loading the second filter. A filter is a circuit that allows spectral components of a signal within a certain bandwidth to pass to its output while attenuating other components of the signal outside of the bandwidth. If a signal is thought of as a mixture, a filter can be thought of as a means by wich to separate desirable components and discard undesirable portions of a signal. An RC filter uses resistors and capacitors to achieve this separation.

The first filter we will look at is the second order low pass filter formed by cascading 2 RC low pass filters:

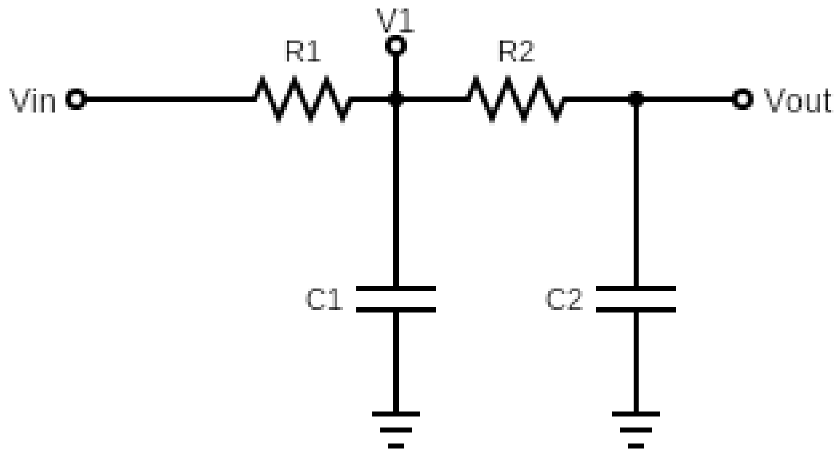

Figure 1. Second Order RC Lowpass Filter

The above circuit, figure 1, is the schematic for the second order low pass filter. The Capacitors act as a means to short high frequency signals/signal components to ground while allowing low frequency components to pass to the output. Note that there are 3 nodes of interest in this circuit: The input V_{in}, V_1, and the output V_{out}. These nodes will be of direct interest for our derivation of the transfer function of this circuit. The resistors R_1 and R_2, as well as, the capacitors C_1 and C_2 are used to tune the bandwidth and the damping/quality factor of the circuit.